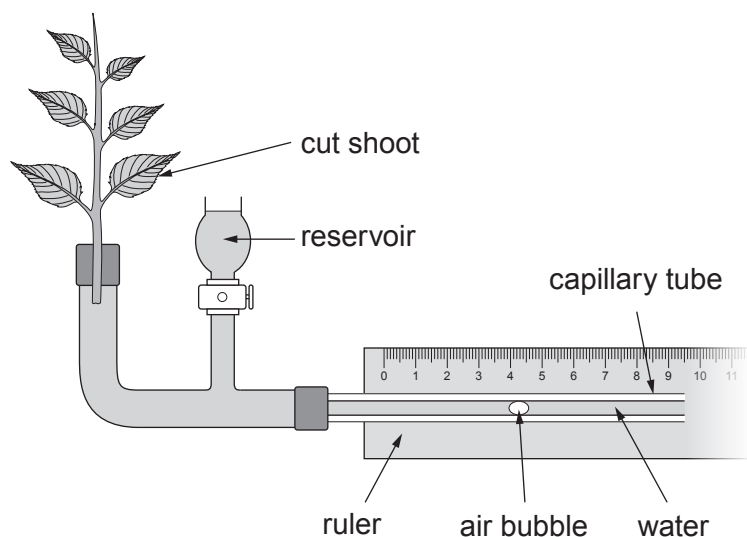


4. A student investigated the effect of wind speed on the rate of transpiration. He set up the apparatus as shown in **Image 4.1**.

Image 4.1



- (a) (i) State the name of the apparatus shown in **Image 4.1**. [1]

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- (ii) When setting up the apparatus, it is important that air bubbles are prevented from entering the xylem.
Describe how the leafy shoot is prepared and the apparatus assembled to avoid this. [2]

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- (iii) State **two environmental** factors that would need to be controlled during this investigation. [1]

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(b) The student then placed a fan at different distances from the apparatus and recorded the time taken for the air bubble to travel 200 mm. The results are shown in **Table 4.2**.

Table 4.2

Distance between the fan and the apparatus/cm	Time taken for the air bubble to travel 200 mm/seconds				Mean rate of water uptake by the leafy shoot/mm ³ min ⁻¹
	Trial 1	Trial 2	Trial 3	mean	
20	168	156	132	152	62.0
40	172	166	145	161	58.5
60	184	170	165	173	54.5
80	188	185	179	184	51.2
100	195	191	190	192	

- (i) I. The internal diameter of the capillary tube was 1 mm. Use the equation below to calculate the volume of water taken up by the shoot when the bubble travels 200 mm. [2]

Volume of a cylinder: $V = \pi r^2 h$

Where
h = distance travelled by the air bubble
 $\pi = 3.14$
r = radius of capillary tube

Volume = mm³

- II. Use your answer to (b)(i) part I and the data in **Table 4.2** to calculate the mean rate of water uptake by the shoot at a fan distance of 100 cm. **Give your answer in mm³ min⁻¹ and write your answer in Table 4.2.** [3]

Space for working



(ii) Describe and explain the results shown in **Table 4.2**.

[3]

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(c) The apparatus shown in **Image 4.1** measures the rate of water uptake.

Give **one** reason why this may not be equal to the rate of water lost in transpiration. [1]

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